

SWEN90010 - High Integrity Systems Engineering

Invariants

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INVARIANTS

While Loop Rule

$$\frac{\{P \wedge B\} S \{P\} \quad P \wedge \neg B \Rightarrow Q}{\{P\} \textbf{while } B \textbf{ do } S \textbf{ done } \{Q\}}$$

P is called an **invariant**

Must satisfy 3 properties:

1. Holds when the loop is encountered
2. Holds after each loop iteration
3. Implies the postcondition when the guard is false

Example

```
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done
```

What does this program compute?

```
result = A * B;
```

Example

```
{true}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Let's prove this

Example

{true}

{?P}

a = A;

{?P1}

result = 0;

{?I}

while a > 0 **do**

 result = result + B;

 a = a - 1;

done

{result = A * B}

(By sequence and
consequence rules)

Example

{true}

{?P}

a = A;

{?P1}

result = 0;

{?I}

(By iteration/while loop
rule)

while a > 0 **do**

{?I $\wedge$ a > 0}

result = result + B;

a = a - 1;

{?I}

done

{?I $\wedge$ a $\leq$ 0}

{result = A * B}

Choosing an Invariant

$$\frac{\{P \wedge B\} S \{P\} \quad P \wedge \neg B \Rightarrow Q}{\{P\} \text{ while } B \text{ do } S \text{ done } \{Q\}}$$

Invariant P must satisfy 3 properties:

1. Holds when the loop is encountered
2. Holds after each loop iteration
3. Implies the postcondition when the guard is false

Usually must talk about the variables modified in the loop

Usually must talk about things in the postcondition

Example 1

```
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done
```

Iter	a	result
0		
1		
2		
3		

Example 1

```
{true}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Iter	a	result
0	A	0
1	A - 1	B
2	A - 2	B * 2
3	A - 3	B * 3

Usually must talk about the variables modified in the loop
result, a

Usually must talk about things in the postcondition
A, B



Example 1

```
{true}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Iter	a	result
0	A	0
1	A - 1	B
2	A - 2	B * 2
3	A - 3	B * 3

A statement that relates: result, a, A, B ?

$$\text{result} + (a * B) = A * B$$

Example 1

```
{true}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Iter	a	result	result + (a * B)
0	A	0	$0 + (A * B)$
1	A - 1	B	$B + ((A - 1) * B)$
2	A - 2	B * 2	$B * 2 + ((A - 2) * B)$
3	A - 3	B * 3	$B * 3 + ((A - 3) * B)$

A statement that relates: result, a, A, B ?

$$\text{result} + (a * B) = A * B$$

Back to the Proof



$$\frac{\{P \wedge B\} S \{P\} \quad P \wedge \neg B \Rightarrow Q}{\{P\} \text{ while } B \text{ do } S \text{ done } \{Q\}}$$

$\{\text{result} + (a * B) = A * B\}$

while $a > 0$ **do**

$\text{result} = \text{result} + B;$

$a = a - 1;$

done

$\{\text{result} = A * B\}$

2 Goals:

$\{\text{result} + (a * B) = A * B$
 $\wedge a > 0\}$

$\text{result} = \text{result} + B;$

$a = a - 1;$

$\{\text{result} + (a * B) = A * B\}$

$\text{result} + (a * B) = A * B$
 $\wedge$

$a \leq 0$

$\rightarrow$

$\text{result} = A * B$

Goal 1

$\text{result} + (a * B) = A * B$
 $\wedge$
 $a \leq 0$
 $\rightarrow$
 $\text{result} = A * B$

Does this hold?

NO

Need to prevent **a** being negative

Strengthen the Invariant

Need to prevent **a** being negative

$$\text{result} + (a * B) = A * B \wedge a \geq 0$$

$$\text{result} + (a * B) = A * B \wedge a = 0$$

→

$$\text{result} = A * B$$

Invariant and negated loop guard now imply postcondition

Goal 2

```
{result + (a * B) = A * B ∧ a ≥ 0 ∧ a > 0}  
result = result + B;  
a = a - 1;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Goal 2

```
{result + (a * B) = A * B ∧ a ≥ 0 ∧ a > 0}  
{?R1}  
result = result + B;  
{?R}  
a = a - 1;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Goal 2

$$\{P[E/x]\} x := E \{P\}$$

$\{\text{result} + (a * B) = A * B \wedge a \geq 0 \wedge a > 0\}$
 $\{?R1\}$
 $\text{result} = \text{result} + B;$
 $\{(\text{result} + (a * B) = A * B \wedge a \geq 0)[a-1/a]\}$
 $a = a - 1;$
 $\{\text{result} + (a * B) = A * B \wedge a \geq 0\}$

Goal 2

```
{result + (a * B) = A * B ∧ a ≥ 0 ∧ a > 0}  
{?R1}  
result = result + B;  
{result + ((a-1) * B) = A * B ∧ a-1 ≥ 0}  
a = a - 1;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Goal 2

```
{result + (a * B) = A * B ∧ a ≥ 0 ∧ a > 0}  
{(result + ((a-1) * B) = A * B ∧ a-1 ≥ 0)[result+B/result]}  
result = result + B;  
{result + ((a-1) * B) = A * B ∧ a-1 ≥ 0}  
a = a - 1;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Goal 2

$$\begin{aligned} &\{\text{result} + (a * B) = A * B \wedge a \geq 0 \wedge a > 0\} \\ &\{(\text{result} + B) + ((a-1) * B) = A * B \wedge a-1 \geq 0\} \\ &\text{result} = \text{result} + B; \\ &\{\text{result} + ((a-1) * B) = A * B \wedge a-1 \geq 0\} \\ &a = a - 1; \\ &\{\text{result} + (a * B) = A * B \wedge a \geq 0\} \end{aligned}$$

Holds If:

$$\begin{aligned} &\text{result} + (a * B) = A * B \wedge a \geq 0 \wedge a > 0 \\ &\rightarrow \\ &(\text{result} + B) + ((a-1) * B) = A * B \wedge a-1 \geq 0 \end{aligned}$$

True

Rest of the Program



```
{true}  
{?P}  
a = A;  
{?P1}  
result = 0;  
{I}  
while a > 0 do  
  {I ∧ a > 0}  
  result = result + B;  
  a = a - 1;  
  {I}  
done  
  {I ∧ a ≤ 0}  
  {result = A * B}
```

$$I = \text{result} + (a * B) = A * B \wedge a \geq 0$$

Rest of the Program

```
{true}  
{?P}  
a = A;  
{?P1}  
result = 0;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Rest of the Program

```
{true}  
{?P}  
a = A;  
{(result + (a * B) = A * B ∧ a ≥ 0)[0/result]}  
result = 0;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Rest of the Program

```
{true}  
{?P}  
a = A;  
{(a * B) = A * B ∧ a ≥ 0}  
result = 0;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Rest of the Program

```
{true}  
{((a * B) = A * B ∧ a ≥ 0)[A/a]}  
a = A;  
{(a * B) = A * B ∧ a ≥ 0}  
result = 0;  
{result + (a * B) = A * B ∧ a ≥ 0}
```

Rest of the Program

{true}
{(A * B) = A * B ∧ A ≥ 0}
a = A;
{(a * B) = A * B ∧ a ≥ 0}
result = 0;
{result + (a * B) = A * B ∧ a ≥ 0}

Holds if:

true =>
((A * B) = A * B ∧ A ≥ 0)

DOESN'T HOLD

Strengthen Precondition

```
{true}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Strengthen Precondition

```
{A ≥ 0}  
a = A;  
result = 0;  
while a > 0 do  
    result = result + B;  
    a = a - 1;  
done  
{result = A * B}
```

Strengthen Precondition



$\{A \geq 0\}$

$\{(A * B) = A * B \wedge A \geq 0\}$

$a = A;$

$\{(a * B) = A * B \wedge a \geq 0\}$

$\text{result} = 0;$

$\{I\}$

while $a > 0$ **do**

$\{I \wedge a > 0\}$

$\text{result} = \text{result} + B;$

$\{\text{result} + ((a-1) * B) = A * B \wedge a-1 \geq 0\}$

$a = a - 1;$

$\{I\}$

done

$\{I \wedge a \leq 0\}$

$\{\text{result} = A * B\}$

$I = \text{result} + (a * B) = A * B \wedge a \geq 0$

QED

Example 2

```
{true}  
i = 0;  
sum = 0;  
while i != N do  
    sum = sum + A[i];  
    i = i + 1;  
done  
{sum =  $\sum_{j=0}^{N-1} A[j]$ }
```

Example 2

{true}

{?Q}

i = 0;

{?P}

sum = 0;

{?I}

while i != N **do**

 sum = sum + A[i];

 i = i + 1;

done

{sum = $\sum_{j=0}^{N-1} A[j]$ }

(By sequence and
consequence rules)

Example 2

{true}

{?Q}

i = 0;

{?P}

(By iteration/while loop
rule)

sum = 0;

{?I}

while i != N **do**

{?I ∧ i != N}

sum = sum + A[i];

i = i + 1;

{?I}

done

{?I ∧ i = N}

{sum = $\sum_{j=0}^{N-1} A[j]$ }

Example 2

Before the loop (0th iteration): $\text{sum} = 0$

After First Iteration: $\text{sum} = A[0]$, $i = 1$

After Second Iteration: $\text{sum} = A[0] + A[1]$, $i = 2$

After Third Iteration: $\text{sum} = A[0] + A[1] + A[2]$, $i = 3$

...

Invariant: $\text{sum} = \sum_{j=0}^{i-1} A[j]$

Postcondition: $\text{sum} = \sum_{j=0}^{N-1} A[j]$



**THE REST IS LEFT AS AN
EXERCISE FOR THE CURIOUS**